(a) $\sum f(x)$ (b) $\sum[x+f(x)]$

(c) $\sum f(x)+x$ (d) $\sum xp(x)$

8. Which of the following is not possible in probability distribution?

(a) $\sum p(x)\geq 0$ (b) $\sum p(x)=1$

(c) $\sum x\,p(x)=2$ (d) $p(x)=-0.5$

9. If c is a constant, then $E(c)$ is

(a) 0 (b) 1

(c) $c\,f(c)$ (d) c

10. A discrete probability distribution may be represented by

(a) table

(b) graph

(c) mathematical equation

(d) all of these

11. A probability density function may be represented by

(a) table

(b) graph

(c) mathematical equation

(d) both (b) and (c)

12. If c is a constant in a continuous probability distribution, then $p(x=c)$ is always equal to

(a) zero (b) one

(c) negative (d) does not exist

13. $E[X-E(X)]$ is equal to

(a) $E(X)$ (b) $V(X)$

(c) 0 (d) $E(X)-X$

14. $E[X-E(X)]^2$ is

(a) $E(X)$ (b) $E(X^2)$

(c) $V(X)$ (d) $S.D(X)$

15. If the random variable takes negative values, then the negative values will have

(a) positive probabilities

(b) negative probabilities

(c) constant probabilities

(d) difficult to tell

16. If we have $f(x)=2x, 0\leq x\leq 1$, then $f(x)$ is a

(a) probability distribution

(b) probability density function

(c) distribution function

(d) continuous random variable

17. $\int_{-\infty}^{\infty} f(x)dx$ is always equal to

(a) zero (b) one

(c) $E(X)$ (d) $f(x)+1$

18. A listing of all the outcomes of an experiment and the probability associated with each outcome is called

(a) probability distribution

(b) probability density function

(c) attributes

(d) distribution function

19. Which one is not an example of random experiment?

(a) A coin is tossed and the outcome is either a head or a tail

(b) A six-sided die is rolled

(c) Some number of persons will be admitted to a hospital emergency room during any hour.

(d) All medical insurance claims received by a company in a given year.

20. A set of numerical values assigned to a sample space is called

(a) random sample

(b) random variable

(c) random numbers

(d) random experiment

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